

Catalytic asymmetric 1,2-Addition/Lactonization tandem reactions for the syntheses of chiral 3-Substituted phthalides using organozinc reagents

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Using chiral phosphoramidate ligand **2d**-Zn (II) complex derived from (1*R*,2*R*)-1,2-diphenylethylenediamine as the catalyst, we have developed efficient catalytic asymmetric 1,2-addition/lactonization tandem reactions of diverse organozinc reagents with varied methyl 2-formylbenzoates for the construction of optically enriched 3-aryl or alkyl substituted phthalides, which are significant building blocks of many important chiral pharmaceuticals and natural products. The corresponding products could be afforded with good to excellent yields (up to 95%) and moderate to good enantioselectivities (up to 89%).

KEYWORDS

addition, catalysis, lactonization, phthalide, tandem reaction

1 | INTRODUCTION

Chiral 3-substituted phthalides (1(3*H*)-isobenzofuranone) are significant backbones of many biologically natural products and medications.^[1] For example, (S)-3-butylphthalide (Figure 1, **I**) is a very useful natural product with wonderful curative effect for the treatment of stroke.^[2] In addition, it shows anticonvulsant action, cerebral anti-ischemic action and so on.^[3] Penicidone A (Figure 1, **II**) was found to exhibit moderate cytotoxicity against some tumor cell lines.^[4] (+)-3-(4'-Methoxybenzyl)-5,7-dimethoxyphthalide (Figure 1, **III**) exhibits cytotoxic activity against human promyelocytic

leukemia (HL-60).^[5] Alcyopterosin E (Figure 1, **IV**) possesses cytotoxic activity.^[6] Hence, developing efficient synthetic methodology for the synthesis of optically chiral 3-substituted phthalides is of great significance.

Despite that a variety of synthetic approaches to 3-substituted phthalides have been developed in the past decades,^[1,7,8] asymmetric synthetic approaches to them are limited.^[1] Among these reported asymmetric synthetic methodologies, chiral auxiliaries were used in most cases^[9] and only a handful of examples using catalytic asymmetric methods were disclosed.^[1,10–20] In 1990, Noyori and coworkers reported their pioneer work on preparing chiral 3-substituted phthalides via catalytic

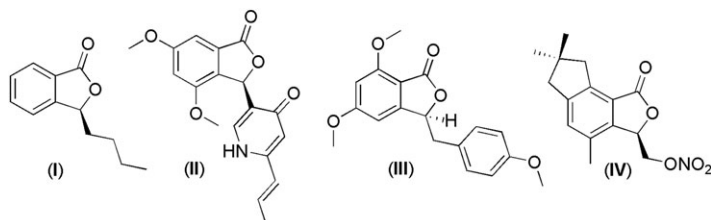


FIGURE 1 Structures of selected chiral 3-substituted phthalides

asymmetric hydrogenation/lactonization tandem reactions.^[11] Lin group^[12] also reported chiral Rh (II) complexes as catalysts in a tandem process obtaining excellent yields and enantioselectivities. In addition, (*S,S*)-dipamp/Co (II) catalyzed cyclization,^[13] Rh(I)-catalyzed one-pot transesterification and [2 + 2 + 2] cycloaddition,^[14] and ProPhenol-catalyzed asymmetric alkynylation followed by a Pd-catalyzed spiroketalization^[15] for the syntheses of chiral 3-substituted phthalides were developed. Moreover, Kitayama and coworkers reported enzymatic catalytic asymmetric oxidation/lactonization tandem reactions of *ortho*-alkyl benzoate for the syntheses of chiral 3-substituted phthalides.^[16] The enantioselectivity of the corresponding products could be up to 99%, but the yields were not satisfied. Furthermore, Wang and coworkers developed an excellent example of organocatalytic asymmetric aldol/lactonization reaction of 2-formylbenzoic esters with ketones/aldehydes for constructing enantioenriched 3-substituted phthalides.^[17]

Besides the above mentioned approaches, catalytic asymmetric 1,2-addition/lactonization tandem reactions using diorganozinc reagent with methyl 2-formylbenzoate is a promising transformation.^[18,19] In 2014, Chang and coworkers reported an excellent method on chiral amino alcohol ligand catalyzed one-pot arylation/lactonization tandem reactions of 2-formylbenzoate with *in situ* generated arylzinc reagent. The corresponding 3-aryl substituted chiral phthalides could be obtained with excellent yields and enantioselectivities.^[19] Despite that pioneering studies have been reported on the catalytic asymmetric preparation of 3-substituted chiral phthalides, the development of highly efficient asymmetric catalytic systems with wide substrate scope using readily prepared chiral ligand is still highly desirable.

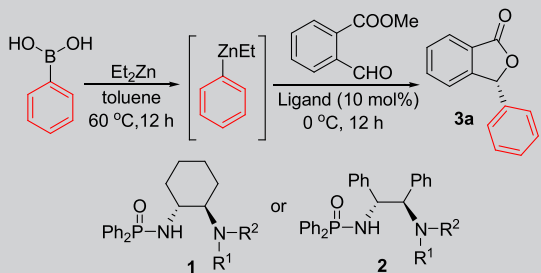
Chiral 1,2-diamino phosphoramidate ligands are very efficient in the organozinc participated catalytic asymmetric reactions,^[20–22] organocatalyzed asymmetric reactions^[23] and enantiodiscrimination of chiral acids.^[24] In our previous study, chiral 1,2-diamino phosphoramidate-Zn (II) complex exhibited excellent effect in the aryl transform addition reactions of aldehydes with *in situ* generated arylalkylzinc reagents.^{21j} As an effort to explore the application of phosphoramidate-Zn (II) catalyzed asymmetric reactions, we herein report an example of catalytic asymmetric 1,2-addition/lactonization tandem reactions

of diorganozinc reagents with methyl 2-formylbenzoates for the construction of 3-substituted chiral phthalides.

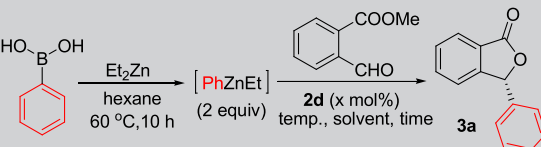
2 | RESULTS AND DISCUSSION

Initially, we used a reaction of catalytic asymmetric 1,2-addition/lactonization of methyl 2-formylbenzoate with *in situ* generated arylzinc reagent using Zn-B exchange method^[25] to screen phosphoramidate chiral ligands. The results are listed in Table 1. When 10 mol% (1*R*,2*R*)-1,2-diaminocyclohexane derived phosphoramidate ligand **1** was used, the desired product of chiral 3-aryl substituted phthalide **3a** was afforded with 88% yield and 11% *ee* (Table 1, Entry 1). The reaction with the use of chiral (1*R*,2*R*)-1,2-diphenylethylenediamine derived phosphoramidate ligand **2a** proceeded to give **3a** with 81% yield and 31% *ee* (Table 1, Entry 2). Chiral (1*R*,2*R*)-1,2-diphenylethylenediamine derived *N,N*-disubstituted phosphoramidate **2b** gave the chiral product **3a** with high yield but only 17% *ee* (Table 1, Entry 3). When *N*-methyl, *N*-isopropyl disubstituted phosphoramidate **2c** was used, the product **3a** was afforded with 89% yield and 63% *ee* (Table 1, Entry 4). Then, a series of *N*-mono substituted phosphoramidate chiral ligands were evaluated (Table 1, Entries 5–8). Chiral phosphoramidate **2d** was found to be the best chiral ligand of choice affording the corresponding product **3a** with 92% yield and 77% *ee* (Table 1, Entry 5).

Next, the reaction conditions, such as solvent, reaction temperature and the loading amount of chiral ligand were optimized. The results are listed in Table 2. Different solvents such as toluene, hexane, NMP and THF were examined and the mixed solvent (toluene/hexane = 1:2) was found to be the best solvent. The resulting product **3a** was afforded with 92% yield and 77% *ee* (Table 2, Entries 1–5). The effect of temperature was studied subsequently. The yield and *ee* increased a little when the reaction temperature was decreased (Table 2, Entries 6). When further decreasing the temperature to –40 °C, chiral product **3a** was obtained with 90% yield and 79% *ee* (Table 2, Entry 7). Increasing the loading amount of chiral ligand to 20 mol%, good enantioselectivity of the product was obtained (82% *ee*, Table 2, Entry 8). When 30 mol% of chiral ligand **2d** was used, 91% yield and

TABLE 1 Screening of chiral 1,2-diamino phosphoramidate ligands in the catalytic asymmetric 1,2-addition/lactonization tandem reaction of methyl 2-formylbenzoate with *in situ* generated arylzinc reagent^a


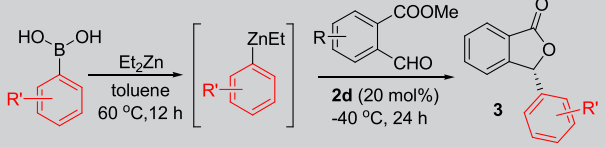
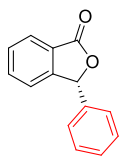
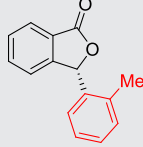
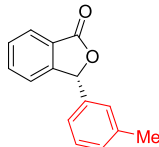
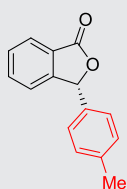
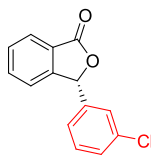
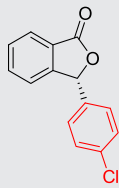
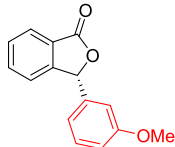
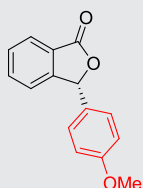
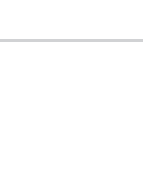
Entry	Ligand	NR ¹ R ²	Yield ^b (%)	ee ^c (%)
1	1	N (CH ₂) ₅	88	11
2	2a	N (CH ₂) ₅	81	31
3	2b	NEt ₂	85	17
4	2c	NMe ⁱ Pr	89	63
5	2d	NH ⁱ Pr	92	77
6	2e	NHCH ₂ Bu	90	70
7	2f	NHCy	91	75
8	2 g	NHCH ₂ Ph	90	69

^aUnless otherwise noted, all reactions were carried out with methyl 2-formylbenzoate in 0.5 mmol scale.^bIsolated yields.^cDetermined by chiral GC analysis. The absolute configuration was assigned by comparison of the sign of specific rotation value with the literature (Lit.8h [α]).^d₂₈ = −39.6 (c 1.0, CHCl₃) for 90% ee (*R*)).**TABLE 2** Optimization of reaction conditions for the catalytic asymmetric 1,2-addition/lactonization tandem reaction of methyl 2-formylbenzoate^a


Entry	x (mol%)	Temp.	Time	Solvent	Yield ^b (%)	ee ^c (%)
1 ^d	10	0 °C	12 h	Toluene/hexane	92	77
2	10	0 °C	12 h	Toluene	90	75
3	10	0 °C	12 h	Hexane	89	74
4	10	0 °C	12 h	Hexane/NMP	trace	N.D. ^d
5	10	0 °C	12 h	Hexane/THF	trace	N.D. ^d
6	10	−20 °C	24 h	Toluene/hexane	92	78
7	10	−40 °C	24 h	Toluene/hexane	90	79
8 ^e	20	−40 °C	24 h	Toluene/hexane	90	82
9	30	−40 °C	24 h	Toluene/hexane	91	82

^aUnless otherwise noted, all reactions were carried out with methyl 2-formylbenzoate in 0.5 mmol scale.^bIsolated yields.^cDetermined by chiral GC analysis. The absolute configuration was assigned by comparison of the sign of specific rotation value with the literature (Lit.8h [α]).^d₂₈ = −39.6 (c 1.0, CHCl₃) for 90% ee (*R*)).^eNot determined.^fThe recovery of **2d** was 95%.

TABLE 3 Catalytic enantioselective 1,2-addition/lactonization tandem reactions of methyl 2-formylbenzoates using **2d** as the chiral ligand^a

				
Entry	Product	Name	Yield ^b (%)	ee ^c (%)
1		(<i>R</i>)- 3a	91	82
2		(<i>R</i>)- 3b	85	76
3		(<i>R</i>)- 3c	95	74
4		(<i>R</i>)- 3d	83	73
5		(<i>R</i>)- 3e	81	87
6		(<i>R</i>)- 3f	82	78
7		(<i>R</i>)- 3 g	85	84
8		(<i>R</i>)- 3 h	84	76
9		(<i>R</i>)- 3i	80	83

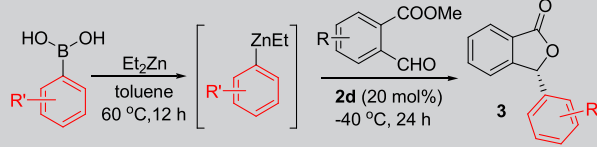
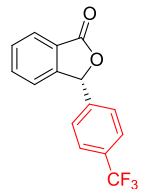
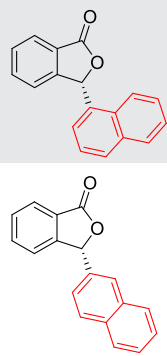
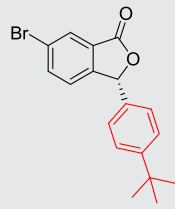
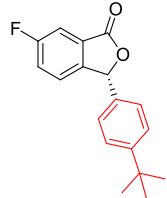
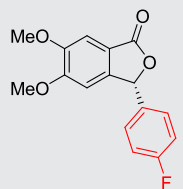
(Continues)

82% *ee* of the product was observed (Table 2, Entry 9). Thus, the reaction was best performed using 2 equivalents of phenyl boronic acid and 10 equivalents of Et_2Zn with 20 mol% chiral ligand **2d** in mixed solvent (toluene/hexane = 1:2) at -40°C for 24 hr. Despite that relatively high loading amount of chiral ligand **2d** (20 mol%)

was required to achieve high level of enantioselectivity, it could be easily recovered by flash column chromatography from the reaction mixture after usual workup.

With the optimized conditions in hand, we investigated the substrate scope and limitation of the catalytic asymmetric tandem reaction system with the use of (1*R*,2*R*)-

TABLE 3 (Continued)

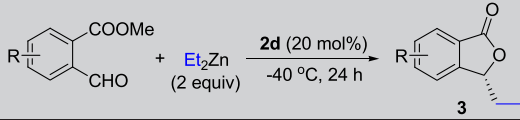
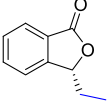
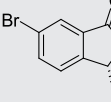
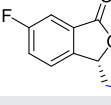
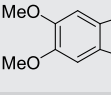
				
Entry	Product	Name	Yield ^b (%)	<i>ee</i> ^c (%)
10		(<i>R</i>)- 3j	86	71
11		(<i>R</i>)- 3k	82	72
12		(<i>R</i>)- 3l	86	73
13		(<i>R</i>)- 3m	83	62
14		(<i>R</i>)- 3n	86	81

^aUnless otherwise noted, all reactions were carried out with methyl 2-formylbenzoate in 0.5 mmol scale.

^bIsolated yields.

^cDetermined by chiral GC or HPLC analysis.

TABLE 4 Catalytic enantioselective 1, 2-addition/lactonization tandem reactions of methyl 2-formylbenzoates with dialkylzinc reagent using **2d** as the chiral ligand^a

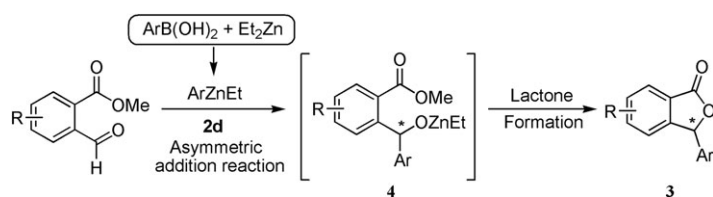
				
Entry	Product	Name	Yield ^b (%)	ee ^c (%)
1		(<i>R</i>)- 3o	92 (90) ^d (86) ^e	89 (86) ^d (87) ^e
2		(<i>R</i>)- 3p	87	82
3		(<i>R</i>)- 3q	82	83
4		(<i>R</i>)- 3r	83	81

^aUnless otherwise noted, all reactions were carried out with methyl 2-formylbenzoate in 0.5 mmol scale.^bIsolated yields.^cDetermined by chiral GC or HPLC analysis.^d10 mol% chiral ligand **2d** was used and the reaction temperature was -20°C .^e1.64 g (10 mmol) of methyl 2-formylbenzoate was used as substrate.

1,2-diphenylethylenediamine derived phosphoramidate ligand **2d**. As shown in Table 3, using phosphoramidate **2d** as the chiral ligand, the catalytic asymmetric 1,2-addition/lactonization tandem reaction of methyl 2-formylbenzoate with a variety of arylzinc reagents *in situ* generated from aromatic boric acids took place, affording the anticipated chiral 3-aryl substituted phthalides in good to excellent yields (80–95% yields) and with good enantioselectivities (73–87% *ee*) (Table 3, Entries 1–9). Aromatic boric acids with electron-withdrawing or electro-donating substituents at the *ortho*-, *meta*-, and *para*-positions of the benzene rings worked well. When 1-naphthylboronic acid and 2-naphthylboronic acid were used as aryl transferring resources, the corresponding products were obtained with moderate yields and enantioselectivities (Table 3, Entries 10 and 11). Some substituted methyl 2-formylbenzoates were examined, and the corresponding products **3 l–n** were all obtained

in good yields with moderate to good *ee* values (Table 3, Entries 12–14). We further tried to use methyl 2-acetylbenzoate as the substrate to react with *in situ* generated ArZnEt species for the construction of oxygen-containing tetrasubstituted stereogenic center, but no product was observed under the optimized reaction conditions owing to the relatively lower activity of the carbonyl group of ketone than that of aldehyde.^[26]

Finally, we tried to use chiral ligand **2d** in the catalytic enantioselective 1,2-addition/lactonization tandem reactions of methyl 2-formylbenzoates with diethylzinc reagent for the synthesis of 3-alkylated chiral phthalide. As shown in Table 4, chiral 3-alkylated phthalide could be afforded with excellent yields (up to 92%) and enantioselectivities (up to 89%). Particularly, a 1-g-scale (10 mmol) preparation of (*R*)-**3o** was established, the corresponding product was obtained in 86% yield with 87% *ee* under optimized reaction conditions (Table 4, Entry 1).

**FIGURE 2** Proposed mechanism for transition zinc-catalyzed asymmetric 1,2-addition/lactone reactions of organozinc reagents with 2-formylbenzoates to access 3-substituted phthalides

In the end, the mechanism was proposed.^{10j, 17, 19} As shown in Figure 2, the active aryethylzinc specie was *in situ* generated by boron to zinc transmetalation. Then it reacted with 2-formylbenzoate to form the diarylmethoxy alkyl zinc intermediate **4** which further generated 3-substituted phthalides **3** via lactone formation.

3 | CONCLUSIONS

In summary, an efficient catalytic asymmetric 1,2-addition/lactonization tandem reaction of organozinc reagents with methyl 2-formylbenzoates using phosphoramidate ligand **2d**-Zn (II) complex as the catalyst was developed. This method provides a direct and convenient way to prepare functionalized 3-aryl or alkyl substituted chiral phthalides with moderate to good enantioselectivities (up to 89%) from readily commercial available arylboronic acids, diorganozinc reagents and methyl 2-formylbenzoates.

4 | EXPERIMENTAL SECTION

4.1 | General Methods

All experiments were carried out in dried glassware with magnetic stirring under an atmosphere of dry nitrogen. ¹H NMR (400 MHz), ¹³C NMR (100 MHz) and ¹⁹F NMR (376 MHz) spectra were recorded in CDCl₃ solutions using a 400 MHz spectrometer. Chemical shifts were reported in parts per million (ppm, δ) relative to CDCl₃ (δ 7.26 for ¹H NMR), or CDCl₃ (δ 77.0 for ¹³C NMR). Multiplicities are indicated as s (singlet), d (doublet), t (triplet), q (quartet), m (multiplet). Commercial reagents were used as received unless otherwise indicated. All solvents were purified and dried prior to use according to standard methods.^[27] Optical rotations were measured on a polarimeter and reported as follows: $[\alpha]_D^{25}$ (c g/100 mL, solvent). The high performance liquid chromatography (HPLC) analysis was performed using Chiralcel AD-H or OD-H column. GC analysis was performed on a gas chromatograph with a FID detector on fused silica chiral capillary column (Chirasil Dex CB column, 25 m Length $\times$ 0.32 mm ID $\times$ 0.25 mm film thickness). Chiral ligands **1**,^{21a} **2a**,^{21e} **2b**,^{21e} **2c**,^{21e} **2d**,^{21e} **2e**,^{21e} **2f**^{21e} and **2g**^{21e} were synthesized according to reported methods.

4.2 | General Procedure For Catalytic Asymmetric Synthesis Of Chiral Phthalide

Under N₂ atmosphere, to a reaction tube which contained corresponding arylboronic acid (1.0 mmol) and toluene (2 ml) was added diethylzinc (5.0 ml, 1.0 M in hexane)

at 0 °C. The solution was stirred for 30 min at room temperature, and then heated at 60 °C for 12 hr. Under N₂ atmosphere, to a solution of chiral ligand **2d** (0.2 equiv.) in toluene (0.5 mL), the above *in situ* generated arylzinc solution was added at 0 °C and stirred for 30 min, then the reaction mixture was cooled to -40 °C and methyl 2-formylbenzoate (0.5 mmol) was then added immediately at -40 °C. The mixture was stirred for 24 hr and quenched by adding aqueous HCl (1.0 M, 5 ml). Extraction with EtOAc (10 ml $\times$ 3) gave combined organic layers that were washed with brine (10 ml), dried over anhydrous Na₂SO₄ and concentrated in vacuo to give a residue which was subjected to silica gel column chromatography (EtOAc/hexane = 1/20), affording the corresponding enantio-enriched 3-substituted phthalide **3**. The *ee* value was determined by Chiral GC or HPLC. The absolute configuration of the phthalide was assigned as (R) by comparison of the optical rotation with reported data.

4.2.1 | 3-Phenylisobenzofuran-1(3H)-one (**3a**)^{8h}

White solid, mp 118–119 °C; 91% yield, 82% *ee*. The *ee* value was determined by Chiral GC Chirasil Dex CB [Column temperature: 165 °C, t_R = 30.88 min (S), t_R = 31.72 min (R)], retention time: 30.88 min (minor) and 31.72 min (major), t_R = 30.88 min (S), t_R = 31.72 min (R); $[\alpha]_D^{31.8}$ = -15.7 (c 0.5, CHCl₃) (Lit.^{8h} $[\alpha]_D^{28}$ = -39.6 (c 1.0, CHCl₃) for 90% *ee* (R)); ¹H NMR (400 MHz, CDCl₃) δ 7.97 (d, J = 7.6 Hz, 1H), 7.65–7.72 (m, 1H), 7.55–7.63 (m, 1H), 7.39–7.43 (m, 3H), 7.33–7.38 (m, 1H), 7.26–7.30 (m, 2H), 6.41 (s, 1H); ¹³C NMR (101 MHz, CDCl₃) δ 170.59, 149.72, 136.43, 134.40, 129.37, 129.01, 127.00, 125.60, 122.93, 82.76.

4.2.2 | 3-(o-Tolyl)isobenzofuran-1(3H)-one (**3b**)^{10k}

White solid, mp 108–109 °C; 85% yield, 76% *ee*. The *ee* value was determined by Chiral HPLC analysis with a Chiralcel AD-H column [(10% IPA/hexane, 1.0 ml/min, 220 nm, retention time: 8.96 min (major) and 9.65 min (minor)), t_R = 8.96 min (R), t_R = 9.65 min (S)]; $[\alpha]_D^{31.6}$ = +17.1 (c 1.0, CHCl₃) (Lit.^{10k} $[\alpha]_D^{22}$ = +41.6 (c 0.62, CHCl₃) for 82% *ee* (R)); ¹H NMR (400 MHz, CDCl₃) δ 7.99 (d, J = 7.6 Hz, 1H), 7.42–7.78 (m, 3H), 7.22–7.41 (m, 2H), 7.09–7.20 (m, 1H), 6.93 (d, J = 7.6 Hz, 1H), 6.70 (s, 1H), 2.52 (s, 3H); ¹³C NMR (100 MHz, CDCl₃) δ 170.65, 149.30, 137.18, 134.21, 134.10, 131.14, 129.38, 129.33, 127.28, 126.43, 125.76, 123.04, 80.55, 19.35.

4.2.3 | 3-(*m*-Tolyl)isobenzofuran-1(3*H*)-one (3c)^{10k}

White solid, mp 85–86 °C; 95% yield, 74% *ee*. The *ee* value was determined by Chiral HPLC analysis with a Chiralcel AD-H column [(10% IPA/hexane, 1.0 ml/min, 220 nm, retention time: 8.63 min (major) and 10.93 min (minor)), $t_R = 8.63$ min (R), $t_R = 10.93$ min (S)]; $[\alpha]_D^{30.8} = -10.2$ (c 1.0, CHCl₃) (Lit.^{10k} $[\alpha]_D^{22} = -40.5$ (c 1.18, CHCl₃) for 90% *ee* (R)); ¹H NMR (400 MHz, CDCl₃) δ 7.99 (d, $J = 7.6$ Hz, 1H), 7.63–7.73 (m, 1H), 7.51–7.62 (m, 1H), 7.34–7.39 (m, 1H), 7.26–7.33 (m, 1H), 7.17–7.24 (m, 1H), 7.06–7.15 (m, 2H), 6.40 (s, 1H), 2.36 (s, 3H); ¹³C NMR (101 MHz, CDCl₃) δ 170.65, 149.83, 138.85, 136.34, 134.34, 130.08, 129.33, 128.86, 127.49, 125.61, 125.55, 124.08, 122.90, 82.84, 21.38.

4.2.4 | 3-(*p*-Tolyl)isobenzofuran-1(3*H*)-one (3d)^{10k}

Light yellow solid, mp 121–122 °C; 83% yield, 73% *ee*. The *ee* value was determined by Chiral HPLC analysis with a Chiralcel AD-H column [(10% IPA/hexane, 1.0 ml/min, 220 nm, retention time: 10.47 min (major) and 15.70 min (minor)), $t_R = 10.47$ min (R), $t_R = 15.70$ min (S)]; $[\alpha]_D^{30.9} = -6.3$ (c 1.0, CHCl₃) (Lit.^{10k} $[\alpha]_D^{22} = -15.2$ (c 0.78, CHCl₃) for 92% *ee* (R)); ¹H NMR (400 MHz, CDCl₃) δ 7.97 (d, $J = 7.5$ Hz, 1H), 7.51–7.75 (m, 2H), 7.34 (d, $J = 7.6$ Hz, 1H), 7.09–7.25 (m, 4H), 6.40 (s, 1H), 2.37 (s, 3H); ¹³C NMR (101 MHz, CDCl₃) δ 170.63, 149.84, 139.36, 134.31, 133.98, 133.40, 129.66, 129.31, 129.08, 127.08, 125.71, 125.69, 125.59, 122.90, 121.77, 82.80, 21.26.

4.2.5 | 3-(3-Chlorophenyl)isobenzofuran-1(3*H*)-one (3e)^{19j}

White solid, mp 116–117 °C; 81% yield, 87% *ee*. The *ee* value was determined by Chiral HPLC analysis with a Chiralcel AD-H column [(10% IPA/hexane, 1.0 ml/min, 220 nm, retention time: 10.34 min (major) and 12.76 min (minor)), $t_R = 10.34$ min (R), $t_R = 12.76$ min (S)]; $[\alpha]_D^{31.4} = -19.7$ (c 1.0, CHCl₃); ¹H NMR (400 MHz, CDCl₃) δ 7.97 (d, $J = 7.6$ Hz, 1H), 7.53–7.79 (m, 2H), 7.02–7.47 (m, 5H), 6.38 (s, 1H); ¹³C NMR (101 MHz, CDCl₃) δ 170.23, 149.06, 138.49, 134.93, 134.60, 130.36, 129.68, 129.49, 126.95, 125.82, 125.35, 125.10, 122.83, 81.70.

4.2.6 | 3-(4-Chlorophenyl)isobenzofuran-1(3*H*)-one (3f)^{10k}

White solid, mp 120–122 °C; 82% yield, 78% *ee*. The *ee* value was determined by Chiral HPLC analysis with a

Chiralcel AD-H column [(10% IPA/hexane, 1.0 ml/min, 220 nm, retention time: 11.98 min (major) and 14.82 min (minor)), $t_R = 11.98$ min (R), $t_R = 14.82$ min (S)]; $[\alpha]_D^{31.1} = -13.5$ (c 1.0, CHCl₃) (Lit.^{10k} $[\alpha]_D^{22} = -32.2$ (c 0.72, CHCl₃) for 88% *ee* (R)); ¹H NMR (400 MHz, CDCl₃) δ 7.99 (d, $J = 7.6$ Hz, 1H), 7.66–7.76 (m, 1H), 7.56–7.63 (m, 1H), 7.31–7.42 (m, 3H), 7.20–7.28 (m, 2H), 6.40 (s, 1H); ¹³C NMR (101 MHz, CDCl₃) δ 170.25, 149.24, 135.33, 134.98, 134.50, 129.61, 129.26, 128.39, 125.82, 125.53, 122.78, 81.86.

4.2.7 | 3-(3-Methoxyphenyl)isobenzofuran-1(3*H*)-one (3 g)^{10k}

White solid, mp 78–80 °C; 85% yield, 84% *ee*. The *ee* value was determined by Chiral HPLC analysis with a Chiralcel AD-H column [(10% IPA/hexane, 1.0 ml/min, 220 nm, retention time: 12.74 min (major) and 17.89 min (minor)), $t_R = 12.74$ min (R), $t_R = 17.89$ min (S)]; $[\alpha]_D^{31.6} = -16.5$ (c 1.0, CHCl₃) (Lit.^{10k} $[\alpha]_D^{22} = -30.4$ (c 0.68, CHCl₃) for 87% *ee* (R)); ¹H NMR (400 MHz, CDCl₃) δ 7.98 (d, $J = 7.6$ Hz, 1H), 7.56–7.78 (m, 2H), 7.23–7.49 (m, 2H), 6.75–7.04 (m, 3H), 6.39 (s, 1H), 3.79 (s, 3H); ¹³C NMR (101 MHz, CDCl₃) δ 170.56, 160.01, 149.62, 137.93, 134.37, 130.09, 129.41, 125.68, 125.47, 122.86, 119.12, 114.65, 112.41, 82.55, 55.34.

4.2.8 | 3-(4-Methoxyphenyl)isobenzofuran-1(3*H*)-one (3 h)^{10k}

White solid, mp 126–127 °C; 84% yield, 76% *ee*. The *ee* value was determined by Chiral HPLC analysis with a Chiralcel AD-H column [(10% IPA/hexane, 1.0 ml/min, 220 nm, retention time: 16.13 min (major) and 21.43 min (minor)), $t_R = 16.13$ min (R), $t_R = 21.43$ min (S)]; $[\alpha]_D^{31.3} = +21.3$ (c 0.5, CHCl₃) (Lit.^{10k} $[\alpha]_D^{22} = +31$ (c 0.42, CHCl₃) for 98% *ee* (R)); ¹H NMR (400 MHz, CDCl₃) δ 7.99 (d, $J = 7.6$ Hz, 1H), 7.53–7.80 (m, 2H), 7.13–7.40 (m, 3H), 6.85–6.96 (m, 2H), 6.40 (s, 1H), 3.82 (s, 3H); ¹³C NMR (101 MHz, CDCl₃) δ 170.56, 160.42, 149.77, 134.27, 129.32, 128.82, 128.29, 125.59, 122.96, 114.34, 82.76, 55.37.

4.2.9 | 3-(4-(Trifluoromethyl)phenyl)isobenzofuran-1(3*H*)-one (3i)^{15j}

Light yellow solid, mp 101–102 °C; 80% yield, 83% *ee*. The *ee* value was determined by Chiral HPLC analysis with a Chiralcel AD-H column [(10% IPA/hexane, 1.0 ml/min, 220 nm, retention time: 9.98 min (major) and 11.10 min (minor)), $t_R = 9.98$ min (R), $t_R = 11.10$ min (S)]; $[\alpha]_D^{31.4} = -19.8$ (c 1.0, CHCl₃); ¹H NMR (400 MHz, CDCl₃) δ 8.01 (d, $J = 7.6$ Hz, 1H), 7.56–7.77 (m, 4H), 7.46 (d,

$J = 8.0$ Hz, 2H), 7.76 (d, $J = 7.6$ Hz, 1H), 6.48 (s, 1H); ^{13}C NMR (101 MHz, CDCl_3) δ 170.17, 148.97, 140.48, 134.62, 131.62, 131.30, 129.77, 127.11, 126.07, 126.35, 125.98, 125.30, 122.69, 122.40, 81.58; ^{19}F NMR (376 MHz, CDCl_3) δ -62.77 (s).

4.2.10 | 3-(Naphthalen-1-yl)isobenzofuran-1(3H)-one (3j)^{9c}

Light yellow solid, mp 174–176 °C; 86% yield, 71% *ee*. The *ee* value was determined by Chiral HPLC analysis with a Chiralcel AD-H column [(10% IPA/hexane, 1.0 ml/min, 220 nm, retention time: 12.71 min (major) and 13.92 min (minor)), $t_R = 12.71$ min (R), $t_R = 13.92$ min (S)]; $[\alpha]_D^{30.7} = -34.3$ (c 1.0, CHCl_3); ^1H NMR (400 MHz, CDCl_3) δ 8.02 (d, $J = 7.6$ Hz, 1H), 7.80–7.98 (m, 4H), 7.50–7.77 (m, 4H), 7.36 (d, $J = 7.6$ Hz, 1H), 7.22–7.30 (m, 1H), 6.58 (s, 1H); ^{13}C NMR (101 MHz, CDCl_3) δ 170.67, 149.75, 134.45, 133.71, 133.61, 133.09, 129.49, 129.19, 128.13, 127.84, 126.87, 126.74, 125.74, 125.63, 123.81, 122.97, 83.00.

4.2.11 | 3-(Naphthalen-2-yl)isobenzofuran-1(3H)-one (3 k)^{10j}

White solid, mp 156–157 °C; 82% yield, 72% *ee*. The *ee* value was determined by Chiral HPLC analysis with a Chiralcel AD-H column [(10% IPA/hexane, 1.0 ml/min, 220 nm, retention time: 14.59 min (major) and 16.42 min (minor)), $t_R = 14.59$ min (R), $t_R = 16.42$ min (S)]; $[\alpha]_D^{31.8} = -6.3$ (c 1.0, CHCl_3); ^1H NMR (400 MHz, CDCl_3) δ 8.03 (d, $J = 7.6$ Hz, 1H), 7.82–7.95 (m, 4H), 7.50–7.71 (m, 4H), 7.37 (d, $J = 7.6$ Hz, 1H), 7.21–7.31 (m, 1H), 6.59 (s, 1H); ^{13}C NMR (101 MHz, CDCl_3) δ 170.64, 149.75, 134.42, 133.69, 133.61, 133.10, 129.48, 129.12, 128.11, 127.84, 126.86, 126.74, 125.75, 125.65, 123.81, 122.96, 82.96, 77.37, 77.06, 76.74.

4.2.12 | 3-(4-Tert-butylphenyl)-6-bromoisobenzofuran-1(3H)-one (3 l)

White solid, mp 101–102 °C; 86% yield, 73% *ee*. The *ee* value was determined by Chiral HPLC analysis with a Chiralcel OD-H column [(6% IPA/hexane, 1.0 ml/min, 220 nm, retention time: 8.31 min (minor) and 9.08 min (major)), $t_R = 8.31$ min (S), $t_R = 9.08$ min (R)]; $[\alpha]_D^{22.6} = -4.4$ (c 1.0, CHCl_3); ^1H NMR (400 MHz, CDCl_3) δ 8.03–8.17 (m, 1H), 7.72–7.82 (m, 1H), 7.36–7.48 (m, 2H), 7.13–7.33 (m, 3H), 6.38 (s, 1H), 1.32 (s, 9H); ^{13}C NMR (101 MHz, CDCl_3) δ 168.9, 152.8, 148.4, 137.3, 132.6, 128.6, 127.9, 126.9, 126.1, 124.6, 123.3, 82.7, 34.7, 31.2.

4.2.13 | 3-(4-Tert-butylphenyl)-6-fluoroisobenzofuran-1(3H)-one (3 m)

White solid, mp 70–72 °C; 83% yield, 62% *ee*. The *ee* value was determined by Chiral HPLC analysis with a Chiralcel OD-H column [(6% IPA/hexane, 1.0 ml/min, 220 nm, retention time: 8.00 min (minor) and 8.59 min (major)), $t_R = 8.00$ min (S), $t_R = 8.59$ min (R)]; $[\alpha]_D^{24.5} = -2.3$ (c 1.0, CHCl_3); ^1H NMR (400 MHz, CDCl_3) δ 7.54–7.66 (m, 1H), 7.30–7.50 (m, 4H), 7.13–7.25 (m, 2H), 6.40 (s, 1H), 1.33 (s, 9H); ^{13}C NMR (101 MHz, CDCl_3) δ 169.4 (d, $J = 3.7$ Hz), 164.5, 162.0, 152.7, 145.3 (d, $J = 2.2$ Hz), 132.9, 127.9, 127.8, 126.4 (d, $J = 84.3$ Hz), 124.8 (d, $J = 8.6$ Hz), 122.3 (d, $J = 23.9$ Hz), 118.8 (d, $J = 23.6$ Hz), 82.6, 34.7, 31.2; ^{19}F NMR (376 MHz, CDCl_3) δ -111.2 (m).

4.2.14 | 3-(4-Fluorophenyl)-5,6-dimethoxyisobenzofuran-1(3H)-one (3n)

White solid, mp 132–134 °C; 86% yield, 81% *ee*. The *ee* value was determined by Chiral HPLC analysis with a Chiralcel AD-H column [(10% IPA/hexane, 1.0 ml/min, 220 nm, retention time: 20.78 min (minor) and 21.85 min (major)), $t_R = 20.78$ min (S), $t_R = 21.85$ min (R)]; $[\alpha]_D^{22.0} = +16.9$ (c 1.0, CHCl_3); ^1H NMR (400 MHz, CDCl_3) δ 7.20–7.39 (m, 3H), 6.98–7.15 (m, 2H), 6.67 (s, 1H), 6.27 (s, 1H), 3.96 (s, 3H), 3.89 (s, 3H); ^{13}C NMR (101 MHz, CDCl_3) δ 170.6, 164.4, 161.9, 132.5 (d, $J = 3.2$ Hz), 129.2 (d, $J = 8.6$ Hz), 117.6, 116.0 (d, $J = 21.9$ Hz), 104.9 (d, $J = 187.6$ Hz), 81.5, 56.43, 56.35; ^{19}F NMR (376 MHz, CDCl_3) δ -111.8 (m).

4.2.15 | 3-Ethylisobenzofuran-1(3H)-one (3o)^[12]

Colorless oil; 92% yield, 89% *ee*. The *ee* value was determined by Chiralcel Dex CB [Column temperature: 130 °C, $t_R = 17.03$ min (R), $t_R = 17.77$ min (S)], retention time: 17.00 min (major) and 17.77 min (minor), $t_R = 17.00$ min (R), $t_R = 17.77$ min (S)]; $[\alpha]_D^{22.1} = +38.4$ (c 1.0, CHCl_3) [Lit.^[12] $[\alpha]_D^{26} = -76$ (c 0.95, CHCl_3) for 97.7% *ee* (S)]; ^1H NMR (400 MHz, CDCl_3) δ 7.85 (d, $J = 7.6$ Hz, 1H), 7.65 (t, $J = 7.5$ Hz, 1H), 7.37–7.58 (m, 2H), 5.35–5.49 (m, 1H), 1.99–2.20 (m, 1H), 1.69–1.87 (m, 1H), 0.96 (t, $J = 7.3$ Hz, 3H); ^{13}C NMR (101 MHz, CDCl_3) δ 170.76, 149.73, 134.02, 129.04, 126.19, 125.54, 121.81, 82.35, 27.59, 8.78.

4.2.16 | 6-Bromo-3-ethylisobenzofuran-1(3H)-one (3p)^[28]

White solid, mp 53–55 °C; 87% yield, 82% *ee*. The *ee* value was determined by Chiralcel Dex CB [Column temperature:

150 °C, $t_R = 23.32$ min (R), $t_R = 24.52$ min (S)], retention time: 23.32 min (major) and 24.52 min (minor)), $t_R = 23.32$ min (R), $t_R = 24.52$ min (S)]; $[\alpha]_D^{24.1} = +25.0$ (c 1.0, CHCl₃); ¹H NMR (400 MHz, CDCl₃) δ 7.95–7.99 (m, 1H), 7.71–7.83 (m, 1H), 7.29–7.43 (m, 1H), 5.35–5.45 (m, 1H), 2.02–2.19 (m, 1H), 1.74–1.90 (m, 1H), 0.97 (t, $J = 7.38$ Hz, 3H); ¹³C NMR (101 MHz, CDCl₃) δ 169.0, 148.4, 137.0, 128.6, 128.4, 123.5, 123.0, 82.3, 27.5, 8.8.

4.2.17 | 3-Ethyl-6-fluoroisobenzofuran-1(3H)-one (3q)

Colorless oil; 82% yield, 83% ee. The ee value was determined by Chirasil Dex CB [Column temperature: 130 °C, $t_R = 15.86$ min (R), $t_R = 16.45$ min (S)], retention time: 15.86 min (major) and 16.45 min (minor)), $t_R = 15.86$ min (R), $t_R = 16.45$ min (S)]; $[\alpha]_D^{21.3} = +23.4$ (c 0.5, CHCl₃); ¹H NMR (400 MHz, CDCl₃) δ 7.51–7.59 (m, 1H), 7.35–7.47 (m, 2H), 5.41–5.50 (m, 1H), 2.00–2.20 (m, 1H), 1.69–1.95 (m, 1H), 1.00 (t, $J = 7.4$ Hz, 3H); ¹³C NMR (101 MHz, CDCl₃) δ 169.5 (d, $J = 4.1$ Hz), 163.1 (d, $J = 249.1$ Hz), 145.2 (d, $J = 2.1$ Hz), 128.3 (d, $J = 8.9$ Hz), 123.4 (d, $J = 8.4$ Hz), 121.9 (d, $J = 8.4$ Hz), 112.0 (d, $J = 23.6$ Hz), 82.2, 27.7, 8.8; ¹⁹F NMR (376 MHz, CDCl₃) δ –111.7 (m).

4.2.18 | Ethyl-5,6-dimethoxyisobenzofuran-1(3H)-one (3r)

White solid, mp 100–101 °C; 83% yield, 81% ee. The ee value was determined by Chiral HPLC analysis with a Chiralcel OD-H column [(4% IPA/hexane, 0.8 ml/min, 220 nm, retention time: 26.94 min (minor) and 30.99 min (major)), $t_R = 26.94$ min (S), $t_R = 30.99$ min (R)]; $[\alpha]_D^{23.6} = +13.4$ (c 1.0, CHCl₃); ¹H NMR (400 MHz, CDCl₃) δ 7.21–7.30 (m, 1H), 6.83 (s, 1H), 5.29–5.39 (m, 1H), 3.97 (s, 3H), 3.93 (s, 3H), 1.97–2.22 (m, 1H), 1.66–1.86 (m, 1H), 0.98 (t, $J = 7.25$ Hz, 3H); ¹³C NMR (101 MHz, CDCl₃) δ 171.0, 154.7, 150.4, 144.3, 118.2, 106.0, 103.1, 81.7, 56.4, 56.3.

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